

S20 Phase II Empirical Validation Roadmap – Detailed Task Breakdown

Overview

Phase II will transition HIMIS-CDI from a design-science specification into an operationally validated governance framework. The roadmap spans 18 months and comprises four workstreams (WS-1 to WS-4), each with defined tasks, deliverables, durations, and measurable success criteria.

WS-1: Real-World Ground-Truth Collection and CDI Threshold Recalibration (Months 1–6)

Task ID	Task Description	Duration	Deliverable	Success Criterion
WS-1-A	Identify partner NHS Trusts (n ≥ 3) and establish data-sharing agreements	2 months	Signed DSA/DPIA documents	Legal approval secured
WS-1-B	Retrospective collection of ≥500 cybersecurity incidents	3 months	Curated incident database	≥500 labelled records
WS-1-C	Blind reassessment using HIMIS-CDI	2 months	HIMIS-CDI scoring outputs	Completion of 500 assessments
WS-1-D	Statistical recalibration of CDI threshold	1 month	Updated threshold analysis	AUROC ≥ 0.70; sensitivity ≥ 0.90
WS-1-E	Documentation and transparency reporting	1 month	Supplementary technical report	Publicly documented methodology

WS-2: External Validation in Live NHS SOC Environments (Months 4–12)

Task ID	Task Description	Duration	Deliverable	Success Criterion
WS-2-A	Integrate HIMIS-CDI into SOC triage workflow	2 months	Integrated operational pipeline	≥2 weeks successful parallel operation
WS-2-B	Prospective CTI evaluation (n ≥ 500)	6 months	Threat-assessment logs	≥500 operational assessments
WS-2-C	Compare HIMIS-CDI recommendations with analyst decisions	2 months	Comparative evaluation report	Non-inferior detection performance
WS-2-D	Revise rubric scoring based on SOC feedback	1 month	Updated governance rubric	$\kappa \geq 0.80$ inter-rater agreement

WS-3: Adversarial Robustness, Fairness Audits, and Concept Drift Monitoring (Months 7–14)

Task ID	Task Description	Duration	Deliverable	Success Criterion
WS-3-A	Adversarial perturbation testing	3 months	Robustness analysis report	No gate flips for perturbations ≤ 0.05
WS-3-B	Fairness audit across NHS Trust categories	3 months	Fairness assessment report	No statistically significant disparities
WS-3-C	Implement concept drift detection	2 months	Drift monitoring module	Detection within 30 days

WS-3-D	Define recalibration periodicity policy	2 months	Governance revision policy	Clinical governance approval
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WS-4: Integration with Live CTI Platforms and OSCAL Alignment (Months 10–18)

Task ID	Task Description	Duration	Deliverable	Success Criterion
WS-4-A	Develop MISP and Anomali ThreatStream connectors	3 months	Operational API plugins	≥10 CTI reports/day ingested
WS-4-B	Align ETP with OSCAL Assessment Results schema	2 months	OSCAL-compliant export module	Validation against OSCAL v1.1
WS-4-C	Develop NHS training materials and workshops	3 months	Training package	≥80% stakeholder satisfaction
WS-4-D	Pilot deployment in second NHS Trust	3 months	Pilot deployment report	Consistent governance outcomes

Conclusion

Phase II is designed to provide the empirical evidence necessary to transition HIMS-CDI from a specification-oriented artefact into an operationally validated governance framework. Collectively, the four workstreams address the principal limitations identified within Section 6.2 (Table 8) of the main manuscript.